

AI Power for COVID-19 Control

Why do patients have to wait for hours to get tested when artificial intelligence (AI) can do it in seconds? Read on to learn how AI and machine learning skills can result in solutions to emerging problems, particularly during a pandemic.

Priya's mother had a fever and a sore throat. Worried that these could be symptoms for COVID-19, she decided to get her mum tested at the nearest public health centre (PHC). When they

reached the PHC, the duo froze for a moment, in shock. There were at least a 100 others, waiting in a queue to be tested as well. It took over six long hours for their turn and to complete the tests. "Wouldn't being exposed to so many people here make us vulnerable to new infections?" Priya asked her mum. "If one wants to get tested, I suppose we have to put up with this," the duo thought. Little did they know that there are AI-powered chatbots that can perform the same preliminary tests,

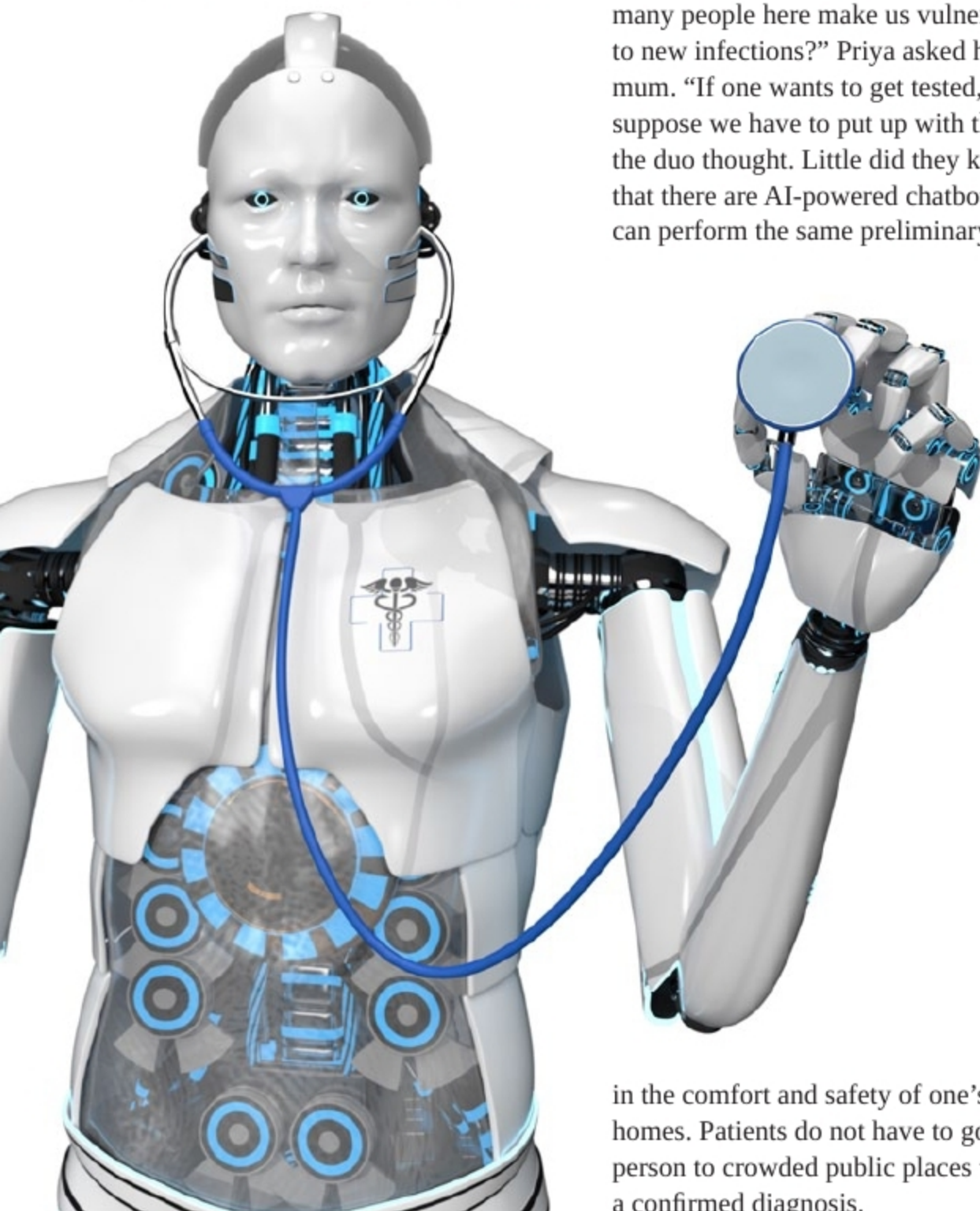
Chatbots for COVID-19 screening

Chatbots are no longer preprogrammed rule-based algorithms that render stock answers. Machine learning (ML), natural language processing (NLP) and artificial intelligence (AI) capabilities are now transforming chatbots, embedding them with a certain degree of 'intelligence' to understand the user's language and search intentions. Deep learning models of Python, such as Universal Sentence Encoders (USE), perform effective text vectorisation so that texts can be understood both for context and semanticity.

For example, let's look at a scenario in which one user asks, "What are the symptoms of COVID-19?" another user types "Corona symptoms", and a third user, "I have a running nose. Is it Corona?" Now, all three questions, though differently worded, have the same meaning. USEs and NLPs understand these language variations and semanticity, and can render direct answers relevant to the user's search intent. Also, chatbots can perform automated preliminary health screenings by deploying forms on websites or mobile applications. Direct answers and test results can be rendered in a matter of seconds.

Neural networks for monitoring progression

Medical imaging like X-rays, CT scans, mammograms, etc, have become fundamental to healthcare monitoring.



in the comfort and safety of one's homes. Patients do not have to go in person to crowded public places to get a confirmed diagnosis.

